Lecture 25

Spinor tensor products and triality

Objectives. *Endomorphisms of spinors form a Clifford algebra, or its even part in odd dimension. As an orthogonal module this is a sum of exterior powers. In type D, chirality and duality decide which exterior powers occur.*

By the end of this lecture you will be able to:

- Identify $\mathrm{End}(S)$ equivariantly with Clifford elements and exterior powers.
- Prove $S \otimes S \cong \bigoplus_{k=0}^{n} \Lambda^k V$ for B_n.
- Prove the D_n rules for $S^{\pm} \otimes S^{\pm}$ and $S^{+} \otimes S^{-}$.
- Decompose the D_4 spinor products and explain triality.

25.1 Clifford algebras as endomorphism algebras

Let V be a complex vector space of dimension N with a nondegenerate symmetric bilinear form q, with the normalisations of Lecture 24. The Clifford algebra $\mathrm{Cl}(V)$ is generated by elements $c(v)$, linear in $v \in V$, with

$$c(u)c(v) + c(v)c(u) = 2q(u,v).$$

The skew maps $A_{uv} \colon V \to V$ are given by

$$A_{uv}(w) = q(v,w)u - q(u,w)v.$$

They span $\mathfrak{so}(V)$ and equal $\tfrac{1}{2}M_{u,v}$ from Lecture 24. The spin homomorphism $\sigma \colon \mathfrak{so}(V) \to \mathrm{Cl}(V)$ and its basic property, proved in Lecture 24, are

$$\sigma(A_{uv}) = \frac{1}{4}[c(u), c(v)], \qquad [\sigma(A), c(w)] = c(Aw). \tag{25.1}$$

The exterior model. Write $N = 2n$ or $N = 2n+1$. Choose $e_1, \dots, e_n, f_1, \dots, f_n \in V$ with

$$q(e_i, e_j) = q(f_i, f_j) = 0, \qquad q(e_i, f_j) = \frac{1}{2}\delta_{ij},$$

and, if N is odd, a vector u_0 orthogonal to them with $q(u_0, u_0) = 1$. Let $W = \mathrm{span}(e_1, \dots, e_n)$ and $S = \Lambda W$, so $\dim S = 2^n$. The algebra homomorphism $\rho_S \colon \mathrm{Cl}(V) \to \mathrm{End}(S)$ (called γ in Lecture 24) is

$$\begin{aligned} \rho_S(c(e_i)) &= e_i \wedge (\cdot), \\ \rho_S(c(f_i)) &= \text{signed contraction removing } e_i, \\ \rho_S(c(u_0)) &= (-1)^{\deg} \qquad (N \text{ odd}). \end{aligned}$$

As in Lecture 24, the subscript names the action on S, because the bare letter ρ is reserved for the Weyl vector. Here the domain of ρ_S is the whole Clifford algebra, and $A \in \mathfrak{so}(V)$ acts on S by $\rho_S(\sigma(A))$.

The conversion table at the end of Lecture 24 compares the normalisations of Lectures 23 to 26. The row of this lecture has the pairing $q(e_j, f_k) = \frac{1}{2}\delta_{jk}$, the isotropic pair $e_j, f_j = \frac{1}{2}(x_j \pm i x_{n+j})$ in an orthonormal basis $x_1, \dots, x_N$, the skew map A_{uv}, and the Clifford map with $\rho_S(c(e_j)) = \mu_j$ and $\rho_S(c(f_j)) = \iota_j$, the operators of Lecture 24.

The Cartan element $H_i = 2A_{e_i f_i}$ acts on V by

$$H_i e_j = \delta_{ij} e_j, \qquad H_i f_j = -\delta_{ij} f_j.$$

On S, Lecture 24 showed that $\rho_S(\sigma(H_i)) = N_i - \frac{1}{2}$, where N_i is 1 on wedges containing e_i and 0 otherwise. So the wedge e_T of the e_t with $t \in T$ has weight

$$\frac{1}{2}(s_1\varepsilon_1 + \cdots + s_n\varepsilon_n), \qquad s_i = +1 \ (i \in T), \quad s_i = -1 \ (i \notin T).$$

We use the positive roots $\varepsilon_i \pm \varepsilon_j$ $(i < j)$, with the ε_i added in type B. Lecture 24 proved:

- for B_n, $n \geq 2$, $S \cong L(\omega_n)$, where $\omega_n = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_n)$;
- for D_n, $n \geq 3$, $S = S^+ \oplus S^-$ with

$$S^+ = \bigoplus_{k \equiv n\,(2)} \Lambda^k W \cong L(\omega_n), \qquad S^- = \bigoplus_{k \equiv n-1\,(2)} \Lambda^k W \cong L(\omega_{n-1}),$$

  where $\omega_{n-1} = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_{n-1} - \varepsilon_n)$.

In the D_n convention, the top wedge $e_1 \wedge \cdots \wedge e_n$ has weight ω_n and degree n. Hence S^+ is the even-degree half only when n is even. Since s has $n - |T|$ minus signs,

$$\text{weights of } S^\pm = \Big\{\frac{1}{2}s : \ s_1 \cdots s_n = \pm 1\Big\}, \qquad \text{each with multiplicity one.}$$

For B_1, the spin operators satisfy

$$\rho_S\sigma(A_{e_1 u_0})1 = \frac{1}{2}e_1, \qquad \rho_S\sigma(A_{f_1 u_0})e_1 = -\frac{1}{2}.$$

Thus S is irreducible, with highest weight $\varepsilon_1/2 = \omega_1$.

An orthonormal basis. Put

$$u_{2j-1} = e_j + f_j, \qquad u_{2j} = i(e_j - f_j) \qquad (1 \leq j \leq n),$$

together with u_0 if N is odd. Then $q(u_a, u_b) = \delta_{ab}$. Let J be the index set, so $|J| = N$, and put $\gamma_a = c(u_a)$. These γ_a are elements of $\mathrm{Cl}(V)$, not values of the map γ of Lecture 24. Thus

$$\gamma_a\gamma_b + \gamma_b\gamma_a = 2\delta_{ab}, \qquad \sigma(A_{u_a u_b}) = \frac{1}{2}\gamma_a\gamma_b \quad (a \neq b).$$

For $I = \{a_1 < \cdots < a_k\} \subseteq J$ put

$$\gamma_I = \gamma_{a_1} \cdots \gamma_{a_k}, \qquad \gamma_\varnothing = 1,$$

and define the spans

$$\mathrm{Cl}_k = \mathrm{span}\{\gamma_I : |I| = k\},$$
$$\mathrm{Cl}^0 = \mathrm{span}\{\gamma_I : |I| \text{ even}\},$$
$$\mathrm{Cl}^1 = \mathrm{span}\{\gamma_I : |I| \text{ odd}\}.$$

Lemma 25.1 (Reordering · *P*). *Let $I, K \subseteq J$ and $a \in J$.*

(i) $\gamma_I \gamma_K = \pm \gamma_{I \triangle K}$, *where $\triangle$ is the symmetric difference.*

(ii) $\mathrm{Cl}(V)$ *is spanned by the γ_I; thus* $\mathrm{Cl}(V) = \mathrm{Cl}^0 + \mathrm{Cl}^1$.

(iii) $\gamma_a \gamma_I \gamma_a = (-1)^{|I|} \gamma_I$ *if $a \notin I$, and* $\gamma_a \gamma_I \gamma_a = (-1)^{|I|-1} \gamma_I$ *if $a \in I$.*

Proof. Distinct γ_a anticommute and $\gamma_a^2 = 1$. Sorting the factors of $\gamma_I \gamma_K$ costs a sign per transposition, and repeated indices cancel. This is (i). The γ_a generate $\mathrm{Cl}(V)$, since c is linear, so (i) gives (ii). For (iii), move the left γ_a through γ_I. It anticommutes with each factor γ_b, $b \neq a$, and commutes with γ_a. Finally, $\gamma_a^2 = 1$. ∎

Reordering makes Cl^0 a subalgebra.

Proposition 25.2 (Clifford algebra and $\mathrm{End}(S)$ · *P*). *(a) If $N = 2n$, then $\rho_S \colon \mathrm{Cl}(V) \to \mathrm{End}(S)$ is an isomorphism, and the γ_I, $I \subseteq J$, form a basis of $\mathrm{Cl}(V)$.*

(b) If $N = 2n + 1$, then ρ_S restricts to an isomorphism $\mathrm{Cl}^0 \to \mathrm{End}(S)$, and the γ_I with $|I|$ even form a basis of Cl^0.

Proof. Let $\mathcal{I}$ be the set of all subsets of J in case (a), and of all even subsets in case (b). In both cases $|\mathcal{I}| = 2^{2n} = \dim \mathrm{End}(S)$, and $\mathcal{I}$ is closed under $\triangle$.

First, $\operatorname{tr} \rho_S(\gamma_I) = 0$ for nonempty $I \in \mathcal{I}$. If $|I|$ is even, choose $a \in I$. If $|I|$ is odd, we are in case (a). Since $|J| = 2n$ is even, $I \neq J$, and we choose $a \notin I$. In both cases Lemma 25.1(iii) gives $\gamma_a \gamma_I \gamma_a = -\gamma_I$. As $\gamma_a^{-1} = \gamma_a$,

$$\operatorname{tr} \rho_S(\gamma_I) = \operatorname{tr}\bigl(\rho_S(\gamma_a)\rho_S(\gamma_I)\rho_S(\gamma_a)^{-1}\bigr) = -\operatorname{tr} \rho_S(\gamma_I).$$

Next, Lemma 25.1(i) gives $\gamma_I \gamma_K = \pm \gamma_{I \triangle K}$ and $\gamma_I \gamma_I = \pm 1$. Hence

$$\operatorname{tr}\bigl(\rho_S(\gamma_I)\rho_S(\gamma_K)\bigr) = \begin{cases} \pm 2^n, & I = K, \\ 0, & I \neq K. \end{cases}$$

Suppose $\sum_{I \in \mathcal{I}} x_I \rho_S(\gamma_I) = 0$. Multiply by $\rho_S(\gamma_K)$ and take traces:

$$0 = \sum_{I \in \mathcal{I}} x_I \operatorname{tr}\bigl(\rho_S(\gamma_I)\rho_S(\gamma_K)\bigr) = \pm 2^n x_K.$$

So the 2^{2n} operators $\rho_S(\gamma_I)$ form a basis of $\mathrm{End}(S)$.

The γ_I, $I \in \mathcal{I}$, span $\mathrm{Cl}(V)$ in case (a) by Lemma 25.1(ii), and span Cl^0 in case (b) by definition. A spanning set with independent images is a basis, and ρ_S maps it bijectively onto a basis. ∎

25.2 Exterior powers and the odd-dimensional square

The algebra $\mathfrak{so}(V)$ acts on $\mathrm{Cl}(V)$ by $X \mapsto [\sigma(A), X]$. This commutator is a derivation of the associative product. By (25.1),

$$[\sigma(A), c(v_1) \cdots c(v_k)] = \sum_{i=1}^{k} c(v_1) \cdots c(Av_i) \cdots c(v_k). \tag{25.2}$$

By Lecture 14, $S \otimes S^* \cong \mathrm{End}(S)$, where A acts on $\mathrm{End}(S)$ by $T \mapsto [\rho_S(\sigma(A)), T]$. As ρ_S is an algebra homomorphism,

$$\rho_S\bigl([\sigma(A), X]\bigr) = [\rho_S(\sigma(A)), \rho_S(X)].$$

Each $\sigma(A)$ is a combination of the $\gamma_a\gamma_b$, so it lies in Cl^0 and its commutator preserves Cl^0. Hence Proposition 25.2 gives equivariant isomorphisms

$$S \otimes S^* \cong \mathrm{End}(S) \cong \begin{cases} \mathrm{Cl}(V), & N = 2n, \\ \mathrm{Cl}^0, & N = 2n+1. \end{cases} \tag{25.3}$$

Proposition 25.3 (Exterior powers in the Clifford algebra · *P*). *For $0 \le k \le N$ define*

$$\varphi_k(v_1 \wedge \cdots \wedge v_k) = \frac{1}{k!} \sum_{\pi \in S_k} \mathrm{sgn}(\pi)\, c(v_{\pi(1)}) \cdots c(v_{\pi(k)}),$$

and let $\Omega = \gamma_J$ be the product of all γ_a.

(a) φ_k is an $\mathfrak{so}(V)$-equivariant isomorphism $\Lambda^k V \to \mathrm{Cl}_k$.

(b) Left multiplication by Ω is an equivariant bijection $\mathrm{Cl}_k \to \mathrm{Cl}_{N-k}$. Hence $\Lambda^k V \cong \Lambda^{N-k} V$.

Proof. The sum is multilinear and alternating in $(v_1, \dots, v_k)$, so φ_k is defined on $\Lambda^k V$. Apply $[\sigma(A), \cdot\,]$ to each term and use (25.2). The result is φ_k of $\sum_i v_1 \wedge \cdots \wedge Av_i \wedge \cdots \wedge v_k$, so φ_k is equivariant. For distinct indices the γ's anticommute, so each reordered product equals $\mathrm{sgn}(\pi)\gamma_I$:

$$\varphi_k(u_{a_1} \wedge \cdots \wedge u_{a_k}) = \gamma_{a_1} \cdots \gamma_{a_k} = \gamma_I.$$

Thus φ_k maps a basis of $\Lambda^k V$ onto the spanning set of Cl_k.

For (b), Lemma 25.1 gives

$$\begin{aligned} \Omega\gamma_I &= \pm\gamma_{J\setminus I} && \text{by (i)}, \\ \gamma_a \Omega \gamma_a &= (-1)^{N-1}\Omega && \text{by (iii)}, \\ \Omega^2 &= \pm 1 && \text{by (i)}. \end{aligned}$$

The first line gives $\Omega\,\mathrm{Cl}_k \subseteq \mathrm{Cl}_{N-k}$. By the second, Ω commutes with each $\gamma_a\gamma_b$, hence with $\sigma(A)$, so left multiplication by Ω commutes with $[\sigma(A), \cdot\,]$. By the third, this map is bijective.

It remains to prove that the γ_I with $|I| = k$ are independent. For $N = 2n$ this is Proposition 25.2(a). For $N = 2n+1$ and k even it is Proposition 25.2(b). For $N = 2n+1$ and k odd, $\gamma_I = \pm\Omega\gamma_{J\setminus I}$ with $|J \setminus I|$ even, and multiplication by Ω is injective. Finally $\varphi_{N-k}^{-1} \circ \Omega \circ \varphi_k$ is an equivariant isomorphism $\Lambda^k V \to \Lambda^{N-k} V$. ∎

Proposition 25.4 (Duals of spin modules · *P*). *(a) For B_n, $S^* \cong S$ for every n.*

(b) For D_n with even $n \ge 4$, $(S^+)^* \cong S^+$ and $(S^-)^* \cong S^-$.

(c) For D_n with odd $n \ge 3$, $(S^+)^* \cong S^-$ and $(S^-)^* \cong S^+$.

Proof. The weights of a dual module are the negatives of the weights, with the same multiplicities. Negating $\frac{1}{2}s$ changes the sign product by

$$(-s_1) \cdots (-s_n) = (-1)^n\, s_1 \cdots s_n.$$

In type B every sign vector occurs once, so $\mathrm{ch}\, S^* = \mathrm{ch}\, S$. In type D, the sign product distinguishes S^+ from S^-. So $\mathrm{ch}(S^\pm)^*$ equals $\mathrm{ch}\, S^\pm$ for n even and $\mathrm{ch}\, S^\mp$ for n odd. Finite-dimensional modules with equal characters are isomorphic, by complete reducibility and highest-weight subtraction (Lecture 17). ∎

Theorem 25.5 (Spinor square for $B_n \cdot P$)**.** *Let $N = 2n+1$ with $n \geq 1$. Then*

$$S \otimes S \cong \mathrm{End}(S) \cong \bigoplus_{k=0}^{n} \Lambda^k V.$$

Proof. By Proposition 25.4(a), $S \otimes S \cong S \otimes S^*$. By (25.3) and Proposition 25.3(a),

$$S \otimes S^* \cong \mathrm{Cl}^0 = \bigoplus_{k \text{ even}} \mathrm{Cl}_k \cong \bigoplus_{\substack{0 \leq k \leq 2n+1 \\ k \text{ even}}} \Lambda^k V.$$

The middle sum is direct since the even γ_I form a basis. For even $k > n$, replace $\Lambda^k V$ by $\Lambda^{2n+1-k}V$, using Proposition 25.3(b). Each $j \in \{0, \dots, n\}$ now occurs once: as $k = j$ if j is even, and from $k = 2n+1-j$ if j is odd. ∎

A natural place to stop. We have identified $\mathrm{End}(S)$ with the Clifford algebra or its even part, and we have proved the spinor square for B_n. The next part treats even dimension, where chirality and duality decide which exterior powers occur.

25.3 Chirality and the D_n rules

From now on $N = 2n$ with $n \geq 1$. Define $S^{\pm}$ by the degree parities above, including $n = 1, 2$. The D_n highest-weight labels require $n \geq 3$. Let $\Gamma \in \mathrm{End}(S)$ act by $+1$ on S^+ and by -1 on S^-. By Proposition 25.2(a) we regard Γ as an element of $\mathrm{Cl}(V)$.

Lemma 25.6 (The chirality element $\cdot$ P)**.** *We have $\Gamma^2 = 1$ and $\Gamma = \lambda\Omega$ for some $\lambda \in \mathbb{C}^\times$. The element Γ commutes with Cl^0 and anticommutes with Cl^1. Left multiplication by Γ is an equivariant bijection $\mathrm{Cl}_k \to \mathrm{Cl}_{2n-k}$.*

Proof. First, $\Gamma^2 = 1$, as Γ acts by ± 1 on $S^{\pm}$. Next, each $\rho_S(c(v))$ is a combination of the operators $e_i \wedge (\cdot)$ and of contractions. These change the degree by ± 1, so $\rho_S(\gamma_a)$ exchanges S^+ and S^-. Since $|J| = 2n$, Lemma 25.1(iii) gives $\gamma_a \Omega \gamma_a = -\Omega$. Hence

$$\Gamma\gamma_a = -\gamma_a\Gamma, \qquad \Omega\gamma_a = -\gamma_a\Omega.$$

Thus $\Gamma\Omega^{-1}$ commutes with every γ_a, hence with $\mathrm{Cl}(V) \cong \mathrm{End}(S)$, and is a nonzero scalar. Since Γ anticommutes with each γ_a, it commutes with even products γ_I and anticommutes with odd ones. The degree statement follows from Proposition 25.3(b), since $\Gamma = \lambda\Omega$. ∎

Left multiplication by Γ preserves Cl_n, and $\Gamma^2 = 1$. Put

$$\mathrm{Cl}_n^{\pm} = \{X \in \mathrm{Cl}_n : \Gamma X = \pm X\}, \qquad \Lambda_{\pm}^n V = \varphi_n^{-1}\big(\mathrm{Cl}_n^{\pm}\big).$$

These are submodules and $\Lambda^n V = \Lambda_+^n V \oplus \Lambda_-^n V$.

The proof of the next theorem studies four blocks of $\mathrm{End}(S)$. We describe its route first, as a preview. The proof does not use this description.

For signs a and b, the block $\mathrm{Hom}(S^a, S^b)$ has source S^a and target S^b. As a module it is $S^b \otimes (S^a)^*$. Even Clifford elements preserve spin parity, so they fill the blocks with $a = b$. Odd Clifford elements exchange spin parity, so they fill the blocks with $a \neq b$.

The element Γ pairs the Clifford degrees k and $2n-k$. For $k < n$, a block keeps one copy from each such pair of the required parity. Left multiplication by Γ preserves the space of degree n, which splits into two eigenspaces of this multiplication.

Finally, duality turns each block into a tensor product. This step depends on the parity of n, by Proposition 25.4. For even n we have $(S^a)^* \cong S^a$. For odd n we have $(S^a)^* \cong S^{-a}$.

Figure 25.1 shows the outcome for $n = 4$ and $n = 3$. For $n = 4$, the tensor squares $S^a \otimes S^a$ are the diagonal blocks. For $n = 3$, they are the off-diagonal blocks.

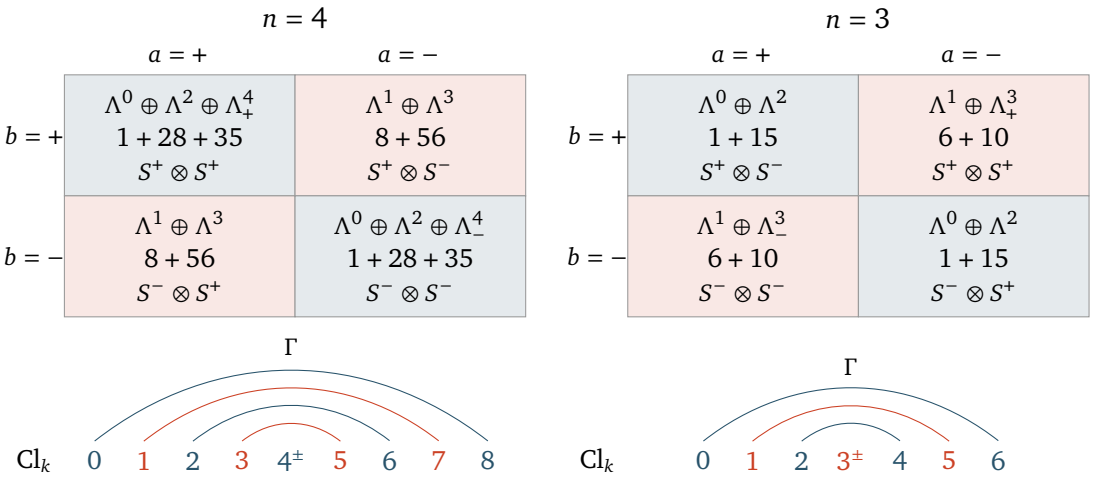


Figure 25.1: The four blocks $\mathrm{Hom}(S^a, S^b)$ of $\mathrm{End}(S)$ for $n = 4$ and $n = 3$. Blue cells hold even Clifford degrees and red cells hold odd ones, and each cell lists the exterior powers of V, their dimensions and the tensor product given by duality. The strips show Γ pairing the degrees k and $2n - k$, with the middle degree splitting into two eigenspaces.

Theorem 25.7 (Chirality rules for $D_n \cdot P$)**.** *Let $N = 2n$ with $n \geq 3$. Then*

$$S^+ \otimes S^+ \cong L(2\omega_n) \oplus \bigoplus_{j \geq 1,\ n-2j \geq 0} \Lambda^{n-2j} V,$$

$$S^- \otimes S^- \cong L(2\omega_{n-1}) \oplus \bigoplus_{j \geq 1,\ n-2j \geq 0} \Lambda^{n-2j} V,$$

$$S^+ \otimes S^- \cong \bigoplus_{j \geq 0,\ n-1-2j \geq 0} \Lambda^{n-1-2j} V.$$

Moreover $\Lambda^n_+ V \cong L(2\omega_n)$ and $\Lambda^n_- V \cong L(2\omega_{n-1})$.

Proof. Step 1: blocks. Put $P_\pm = \frac{1}{2}(1 \pm \Gamma)$, the projections onto $S^\pm$. For signs a, b, the space $\mathrm{Hom}(S^a, S^b)$ is $P_b \,\mathrm{End}(S) P_a$. It is a submodule isomorphic to $S^b \otimes (S^a)^*$, because $\sigma(A) \in \mathrm{Cl}^0$ commutes with Γ. By Lemma 25.6,

$$P_b X P_a = \begin{cases} \delta_{a,b}\, P_b X, & X \in \mathrm{Cl}^0, \\ \delta_{-a,b}\, P_b X, & X \in \mathrm{Cl}^1 . \end{cases}$$

Hence $\mathrm{Hom}(S^a, S^b) = P_b \,\mathrm{Cl}^p$, where $p = 0$ if $a = b$ and $p = 1$ if $a \neq b$.

Step 2: degrees. Write Cl^p as the sum of the $\mathrm{Cl}_k \oplus_\Gamma \mathrm{Cl}_k$ with $k < n$, $k \equiv p$, and of Cl_n if $n \equiv p$. Since $P_b \Gamma = b P_b$,

$$P_b \left(\mathrm{Cl}_k \oplus_\Gamma \mathrm{Cl}_k \right) = P_b \,\mathrm{Cl}_k .$$

For $k < n$ consider

$$\mathrm{Cl}_k \longrightarrow P_b \,\mathrm{Cl}_k, \qquad x \longmapsto P_b x = \frac{1}{2} x + \frac{b}{2} \Gamma x.$$

It is injective, as $\Gamma x \in \mathrm{Cl}_{2n-k}$ and $2n-k \neq k$. It is equivariant, since Γ commutes with each $\sigma(A)$. Also $P_b \,\mathrm{Cl}_n = \mathrm{Cl}_n^b$. The pieces lie in different $\mathrm{Cl}_k \oplus \mathrm{Cl}_{2n-k}$, so

$$\mathrm{Hom}(S^a, S^b) \cong \bigoplus_{k<n,\ k\equiv p} \Lambda^k V \ \oplus \ \begin{cases} \mathrm{Cl}_n^b, & n \equiv p, \\ 0, & n \not\equiv p. \end{cases}$$

Step 3: duals. Put $\eta = (-1)^n$. By Proposition 25.4, $(S^c)^* \cong S^{\eta c}$ for each sign c. Hence

$$S^b \otimes S^b \cong S^b \otimes (S^{\eta b})^* = \mathrm{Hom}(S^{\eta b}, S^b),$$
$$S^+ \otimes S^- \cong S^+ \otimes (S^{-\eta})^* = \mathrm{Hom}(S^{-\eta}, S^+).$$

In the first line $p \equiv n$, since $\eta b = b$ exactly when n is even. In the second line $p \not\equiv n$. Step 2 gives the third formula with $k = n-1-2j$, and

$$S^b \otimes S^b \cong \mathrm{Cl}_n^b \oplus \bigoplus_{k<n,\ k\equiv n} \Lambda^k V.$$

Step 4: the middle degree. By Proposition 25.3(a) and Lecture 23,

$$\mathrm{Cl}_n^+ \oplus \mathrm{Cl}_n^- = \mathrm{Cl}_n \cong \Lambda^n V \cong L(2\omega_{n-1}) \oplus L(2\omega_n).$$

The weights of $S^b \otimes S^b$ are $\frac{1}{2}(s+s')$ with $s_1 \cdots s_n = s'_1 \cdots s'_n = b$. In ε-coordinates the two extreme weights force

$$2\omega_n = (1,\dots,1,1) \implies s = s' = (1,\dots,1,1), \quad b = +1,$$
$$2\omega_{n-1} = (1,\dots,1,-1) \implies s = s' = (1,\dots,1,-1), \quad b = -1.$$

By Step 3, Cl_n^b is isomorphic to a summand of $S^b \otimes S^b$. Hence Cl_n^- contains no $L(2\omega_n)$, and Cl_n^+ contains no $L(2\omega_{n-1})$. Uniqueness of multiplicities gives $\mathrm{Cl}_n^+ \cong L(2\omega_n)$ and $\mathrm{Cl}_n^- \cong L(2\omega_{n-1})$. ∎

The middle degree has now been split twice. Lecture 23 split $\Lambda^n V$ by the Hodge operator $\star$, and the theorem above splits it by Γ. We now compare the two splittings.

Let $x_1, \dots, x_{2n}$ be the orthonormal basis of the conversion table in Lecture 24, so that $u_{2j-1} = x_j$ and $u_{2j} = -x_{n+j}$. Lecture 23 calls this basis u_j. Its volume element and its constant are

$$\mathrm{vol} = x_1 \wedge x_{n+1} \wedge x_2 \wedge x_{n+2} \wedge \cdots \wedge x_n \wedge x_{2n}, \qquad c_n = i^n (-1)^{n(n+1)/2}.$$

Thus $c_n = 1$ for even n, and $c_n = -i$ for odd n.

Proposition 25.8 (Chirality and the Hodge operator · *P*). *Let $N = 2n$ with $n \geq 3$. Let $\Gamma_\Lambda \colon \Lambda^n V \to \Lambda^n V$ be given by $\Gamma_\Lambda = \varphi_n^{-1} \circ (\Gamma\,\cdot) \circ \varphi_n$, and let $\star$ be the Hodge operator of* vol. *Then*

$$\Gamma_\Lambda = c_n^{-1} \star .$$

Hence $\Lambda^n_+ V$ is the c_n-eigenspace of $\star$, and $\Lambda^n_- V$ is its $(-c_n)$-eigenspace. In particular, $\Gamma_\Lambda = \star$ for even n and $\Gamma_\Lambda = i\star$ for odd n.

Proof. The vectors e_j and f_j here are $1/\sqrt{2}$ times the vectors e_j and e_{-j} of Lecture 23. So the two lectures use the same Cartan elements and the same positive roots. The modules $L(2\omega_n)$ and $L(2\omega_{n-1})$ are not isomorphic, and each occurs once in $\Lambda^n V$. So both lectures mean the same two submodules of $\Lambda^n V$.

By Theorem 25.7, Γ_Λ acts by $+1$ on $L(2\omega_n)$ and by -1 on $L(2\omega_{n-1})$. By the proposition on Hodge eigenvalues in Lecture 23, $\star$ acts by c_n and by $-c_n$ on these submodules. Hence $\Gamma_\Lambda = c_n^{-1}\star$ on both of them. ∎

The formula depends on vol, not only on the form. Since $u_{2j} = -x_{n+j}$, we have $\varphi_{2n}(\mathrm{vol}) = (-1)^n\Omega$. Replacing vol by $-$vol negates $\star$, as Lecture 23 notes.

25.4 The D_4 spinor products

Take $n = 4$ and $V = \mathbb{C}^8$. Use the D_4 roots and fundamental weights of Lecture 23. Using Lecture 23 for the exterior powers, write

$$8_v = V = L(\omega_1), \qquad 8_s = S^+ = L(\omega_4), \qquad 8_c = S^- = L(\omega_3),$$
$$28 = \Lambda^2 V = L(\omega_2), \quad 56_v = \Lambda^3 V = L(\omega_3 + \omega_4), \quad 35_s = L(2\omega_4).$$

Theorem 25.7 with $n = 4$ gives

$$8_s \otimes 8_s \cong 35_s \oplus \Lambda^2 V \oplus \Lambda^0 V = 1 \oplus 28 \oplus 35_s,$$
$$8_s \otimes 8_c \cong \Lambda^3 V \oplus \Lambda^1 V = 8_v \oplus 56_v.$$

We confirm both by highest weights. Use the Weyl formula of Lecture 23, with Weyl vector $\rho_W = (3, 2, 1, 0)$, $r = \rho_W$ and $a = \lambda + \rho_W$. The denominator is

$$\prod_{i<j}(r_i^2 - r_j^2) = (9-4)(9-1)(9-0)(4-1)(4-0)(1-0)$$
$$= 5 \cdot 8 \cdot 9 \cdot 3 \cdot 4 \cdot 1 = 4320.$$

Therefore

$$\dim L(2\omega_4) = \frac{7 \cdot 12 \cdot 15 \cdot 5 \cdot 8 \cdot 3}{4320} = \frac{151200}{4320} = 35, \quad a = (4, 3, 2, 1),$$
$$\dim L(\omega_3 + \omega_4) = \frac{7 \cdot 12 \cdot 16 \cdot 5 \cdot 9 \cdot 4}{4320} = \frac{241920}{4320} = 56, \quad a = (4, 3, 2, 0).$$

The square. The weights of $8_s \otimes 8_s$ are $\frac{1}{2}(s + s')$ with $s_1 \cdots s_4 = s'_1 \cdots s'_4 = 1$. A coordinate is ± 1 where s and s' agree and 0 where they differ, and they differ in an even number of places. The dominant weights and multiplicities are

$$(1, 1, 1, 1) : 1, \qquad (1, 1, 0, 0) : 2, \qquad (0, 0, 0, 0) : 8.$$

For $(1, 1, 0, 0)$, $s = (1, 1, x, y)$ and $s' = (1, 1, -x, -y)$ with $xy = 1$.

The weight $(1, 1, 1, 1) = 2\omega_4$ is maximal, so $L(2\omega_4)$ is a summand. Its complement Q satisfies

$$\dim Q = 64 - 35 = 29 > 8 = \dim(8_s \otimes 8_s)_0.$$

So some simple summand of Q has nonzero highest weight. Since $2\omega_4$ has multiplicity one, it is not a weight of Q, so this highest weight is ω_2. Hence $Q \supseteq L(\omega_2)$, of dimension 28. The remaining one-dimensional module is trivial, since $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$.

The mixed product. For $8_s \otimes 8_c$, $s_1 \cdots s_4 = 1$ and $s'_1 \cdots s'_4 = -1$, so s and s' differ in an odd number of places. The dominant weights are

$$(1, 1, 1, 0) : 1, \qquad (1, 0, 0, 0) : 4.$$

The maximal weight $\omega_3 + \omega_4$ gives a summand of dimension 56. The complement has dimension

$$64 - 56 = 8.$$

Its only possible dominant weight is ω_1, since 0 is not a weight and $\omega_3 + \omega_4$ has multiplicity one. So it is $L(\omega_1) = 8_v$.

25.5 Triality

Figure 25.2 shows the D_4 diagram, with its three outer nodes and their modules.

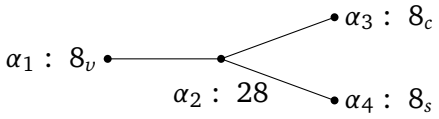


Figure 25.2: The D_4 diagram. Each node shows its simple root and the simple module whose highest weight is the matching fundamental weight.

Every permutation of $\{1, 3, 4\}$ fixing 2 preserves the Cartan matrix.

Theorem 25.9 (Serre's theorem · *C*). *Let $\mathfrak{g}$ be complex semisimple with Cartan matrix $a_{ij} = \alpha_j(h_i)$ and generators $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = [e_i, f_i] = \alpha_i^\vee$. Then $\mathfrak{g}$ is presented by these generators and the Serre relations stated in Lecture 10, whose coefficients are the entries a_{ij}. (Humphreys, Sections 18.3–18.4.)*

If a permutation π satisfies $a_{\pi(i)\pi(j)} = a_{ij}$, the elements $e_{\pi(i)}, f_{\pi(i)}, h_{\pi(i)}$ satisfy these relations. So there is a homomorphism $\tau_\pi : \mathfrak{g} \to \mathfrak{g}$ with $e_i \mapsto e_{\pi(i)}$, $f_i \mapsto f_{\pi(i)}$, $h_i \mapsto h_{\pi(i)}$. Since $\tau_{\pi^{-1}}$ inverts it on generators, τ_π is an automorphism.

Lemma 25.10 (Twisting · *P*). *For modules (M, ρ_M) and $(M', \rho_{M'})$, let M^π be M with x acting by $\rho_M(\tau_\pi^{-1} x)$. Then $(M \otimes M')^\pi = M^\pi \otimes (M')^\pi$, and*

$$L(\lambda)^\pi \cong L(\lambda^\pi), \qquad \lambda^\pi(h_i) = \lambda(h_{\pi^{-1}(i)}).$$

Proof. The tensor action $x \mapsto \rho_M(x) \otimes 1 + 1 \otimes \rho_{M'}(x)$ evaluated at $\tau_\pi^{-1} x$ is the tensor action of the twisted modules. The set of operators is unchanged, so $L(\lambda)^\pi$ is irreducible. Let v be a highest vector of $L(\lambda)$. In $L(\lambda)^\pi$, e_i acts by $e_{\pi^{-1}(i)}$, which sends v to 0. Also h_i acts on v by $\lambda(h_{\pi^{-1}(i)})$. The classification of Lecture 13 gives the isomorphism. ∎

For a fundamental weight the labels are

$$\omega_i^\pi(h_j) = \omega_i(h_{\pi^{-1}(j)}) = \delta_{i,\pi^{-1}(j)} = \delta_{\pi(i),j},$$

so $L(\omega_i)^\pi \cong L(\omega_{\pi(i)})$. Thus the permutations of $\{1, 3, 4\}$ permute the three eight-dimensional modules

$$8_v = L(\omega_1), \qquad 8_c = L(\omega_3), \qquad 8_s = L(\omega_4)$$

transitively. They fix ω_2, so 28 is fixed. This is *triality*.

The twisting lemma also transports tensor decompositions. Twisting by π preserves tensor products and direct sums, and it replaces each highest weight λ by λ^π. Apply this to the two products of § 25.4. The transposition of 1 and 4 sends the decomposition of $8_s \otimes 8_s$ to one of $8_v \otimes 8_v$, and that of $8_s \otimes 8_c$ to one of $8_v \otimes 8_c$. The transposition of 3 and 4 sends $8_s \otimes 8_s$ to $8_c \otimes 8_c$. The permutation with $\pi(4) = 1$, $\pi(3) = 4$ and $\pi(1) = 3$ sends $8_s \otimes 8_c$ to $8_v \otimes 8_s$, and Exercise 25.6 carries out this step. The results are as follows.

product	decomposition	dimensions
$8_v \otimes 8_v$	$1 \oplus 28 \oplus L(2\omega_1)$	$1+28+35$
$8_s \otimes 8_s$	$1 \oplus 28 \oplus L(2\omega_4)$	$1+28+35$
$8_c \otimes 8_c$	$1 \oplus 28 \oplus L(2\omega_3)$	$1+28+35$
$8_s \otimes 8_c$	$8_v \oplus L(\omega_3+\omega_4)$	$8+56$
$8_v \otimes 8_s$	$8_c \oplus L(\omega_1+\omega_4)$	$8+56$
$8_v \otimes 8_c$	$8_s \oplus L(\omega_1+\omega_3)$	$8+56$

These rows are not one product computed again. The six modules are different, and twisting relates them. The first row agrees with Lecture 23, where $V \otimes V = \operatorname{Sym}^2 V \oplus \Lambda^2 V$ has the summands $L(2\omega_1)$, $\mathbb{C}$ and $\Lambda^2 V$.

Self-checks

(1) For D_5, which of $S^+ \otimes S^+$ and $S^+ \otimes S^-$ contains $\mathbb{C}$?

(2) List the summands and dimensions of $S \otimes S$ for B_3.

(3) In D_n, where does the copy of V occur, for n even and for n odd?

Answers. (1) $S^+ \otimes S^- \cong \Lambda^4 V \oplus \Lambda^2 V \oplus \mathbb{C}$, since $(S^+)^* \cong S^-$ for odd n. (2) $\mathbb{C} \oplus V \oplus \Lambda^2 V \oplus \Lambda^3 V$, with $1+7+21+35 = 64 = 8^2$. (3) In $S^+ \otimes S^-$ for n even, and in $S^\pm \otimes S^\pm$ for n odd.

Summary

The algebra $\operatorname{End}(S)$ is the Clifford algebra, or its even part in odd dimension. Hence $S \otimes S^*$ is a sum of exterior powers.

For B_n, S is self-dual and $S \otimes S \cong \bigoplus_{k=0}^{n} \Lambda^k V$. For D_n, the chirality element Γ sorts degrees by parity and splits the middle degree. This splitting is the Hodge splitting of Lecture 23, with $\Gamma_\Lambda = c_n^{-1}\star$. Self-duality of $S^\pm$ depends on the parity of n. In D_4, triality permutes 8_v, 8_s and 8_c, and it transports their tensor products.

Reading

Fulton–Harris, §§20.1–20.3 (octonionic triality in §20.3). Goodman–Wallach, Chapter 6. Humphreys, §§18.3–18.4, for Serre's theorem.